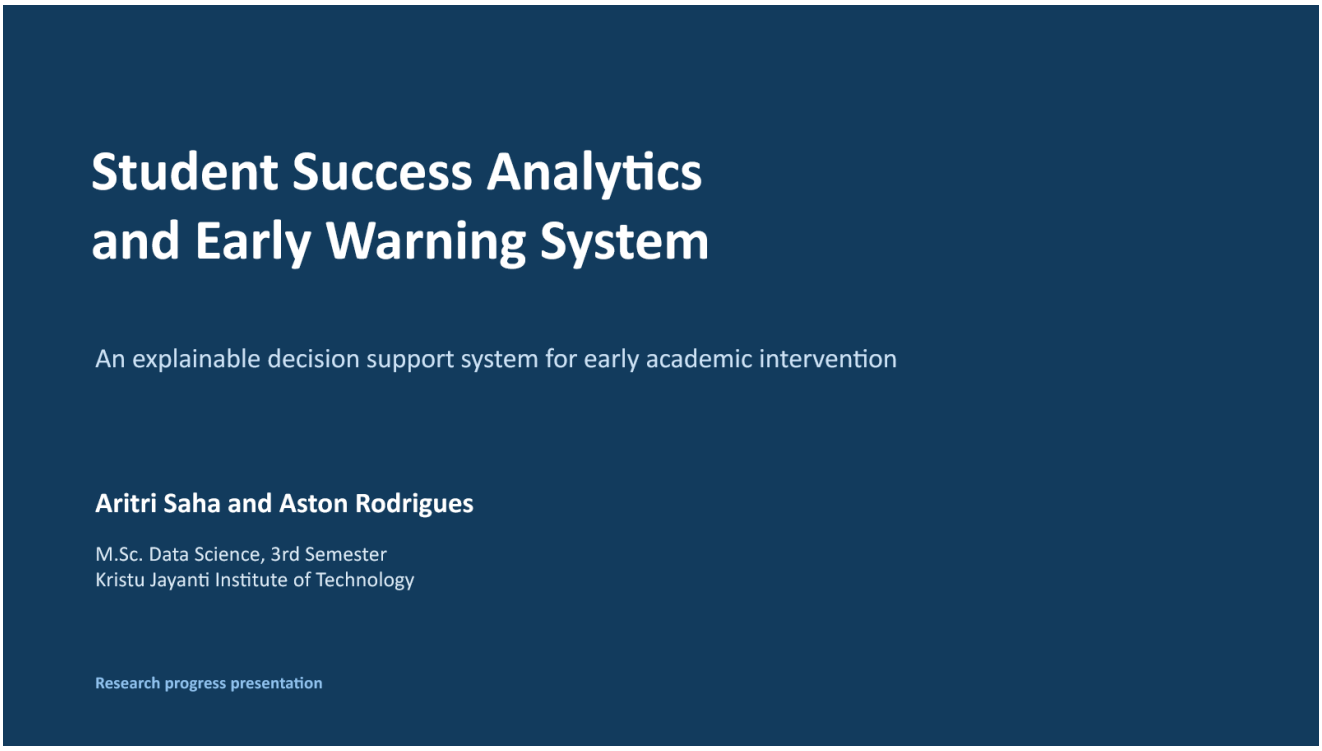
Slide 1: Student Success Analytics and Early Warning System

Speaker: Aritri	Suggested time: 30 seconds

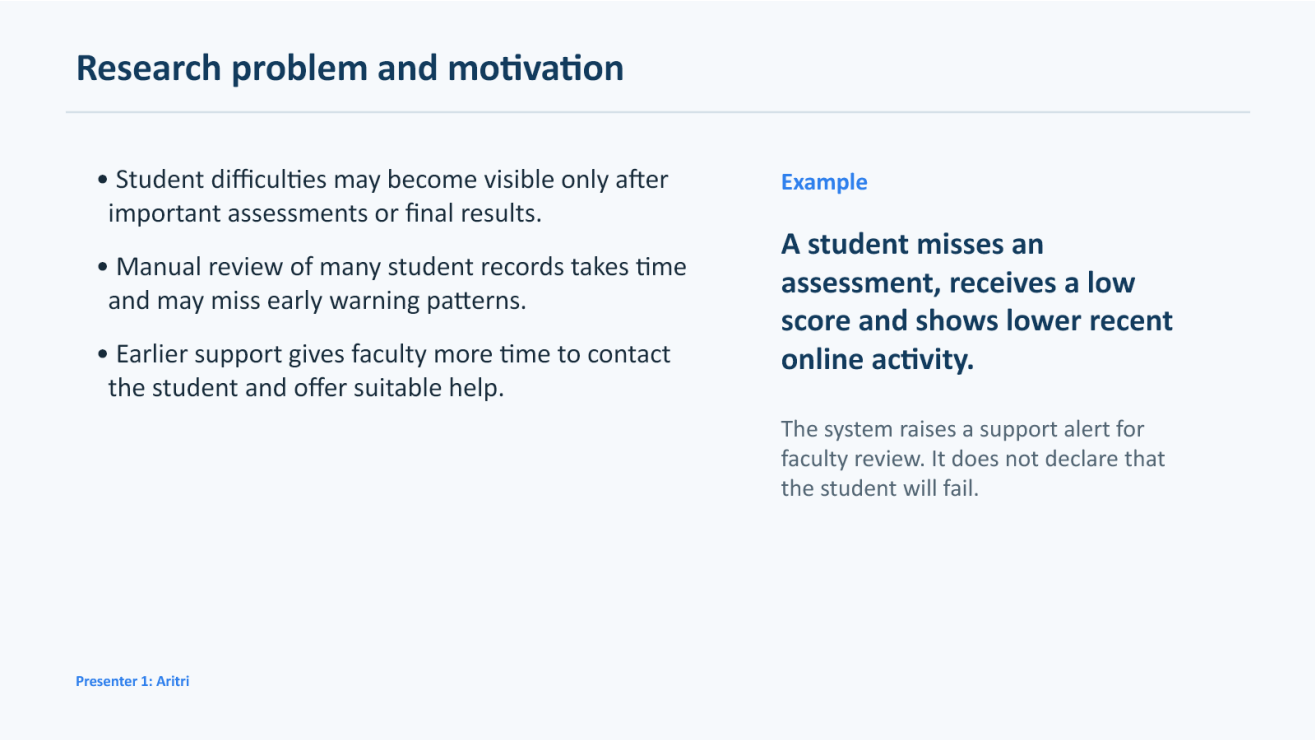


Speaker notes

Good morning. We are Aritri Saha and Aston Rodrigues. Our research examines how early academic and learning activity can help faculty identify students who may need support. The system provides decision support, while faculty remain responsible for every action.

Slide 2: Research problem and motivation

Speaker: Aritri	Suggested time: 50 seconds

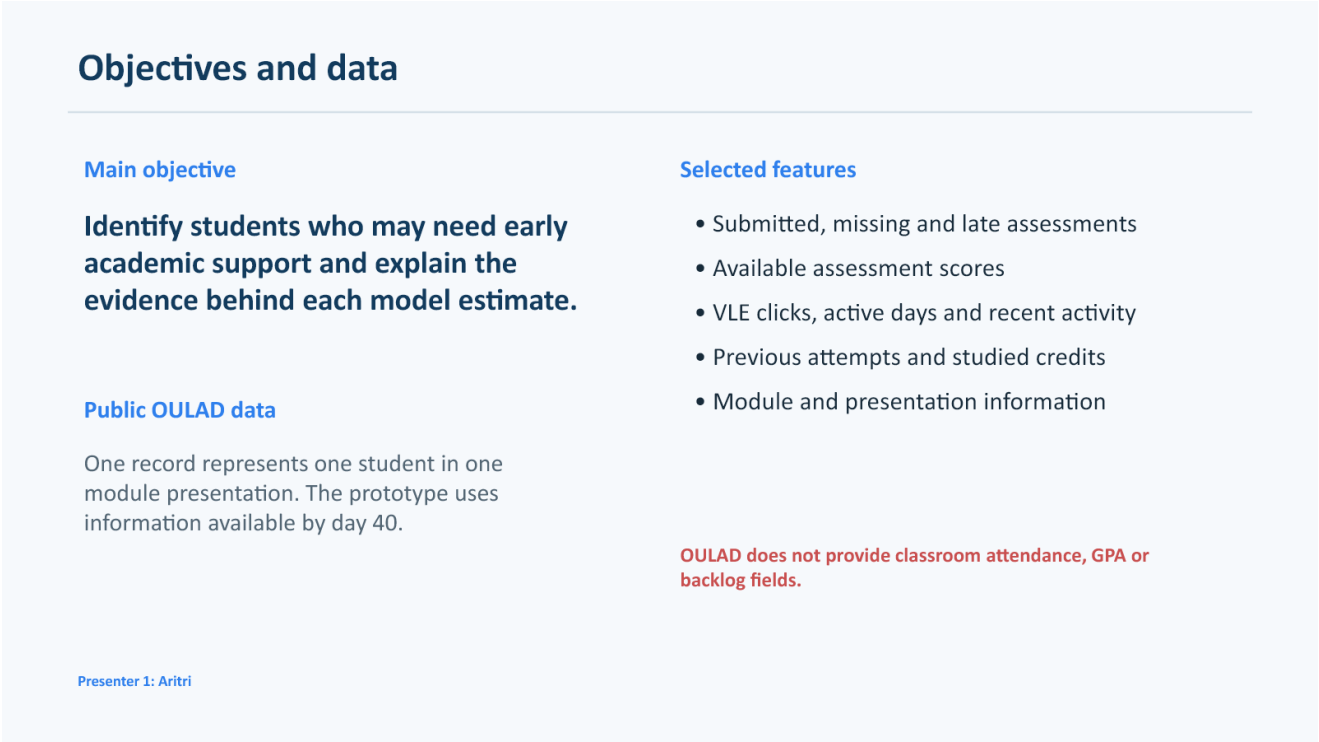


Speaker notes

Many institutions collect marks, submissions and learning-platform activity, but staff may review these records separately. A student can therefore struggle for several weeks before the pattern becomes clear. Our research studies whether an early warning system can organise these signals and help faculty respond sooner. The alert is only an indication for review.

Slide 3: Research objectives and data

Speaker: Aritri	Suggested time: 60 seconds

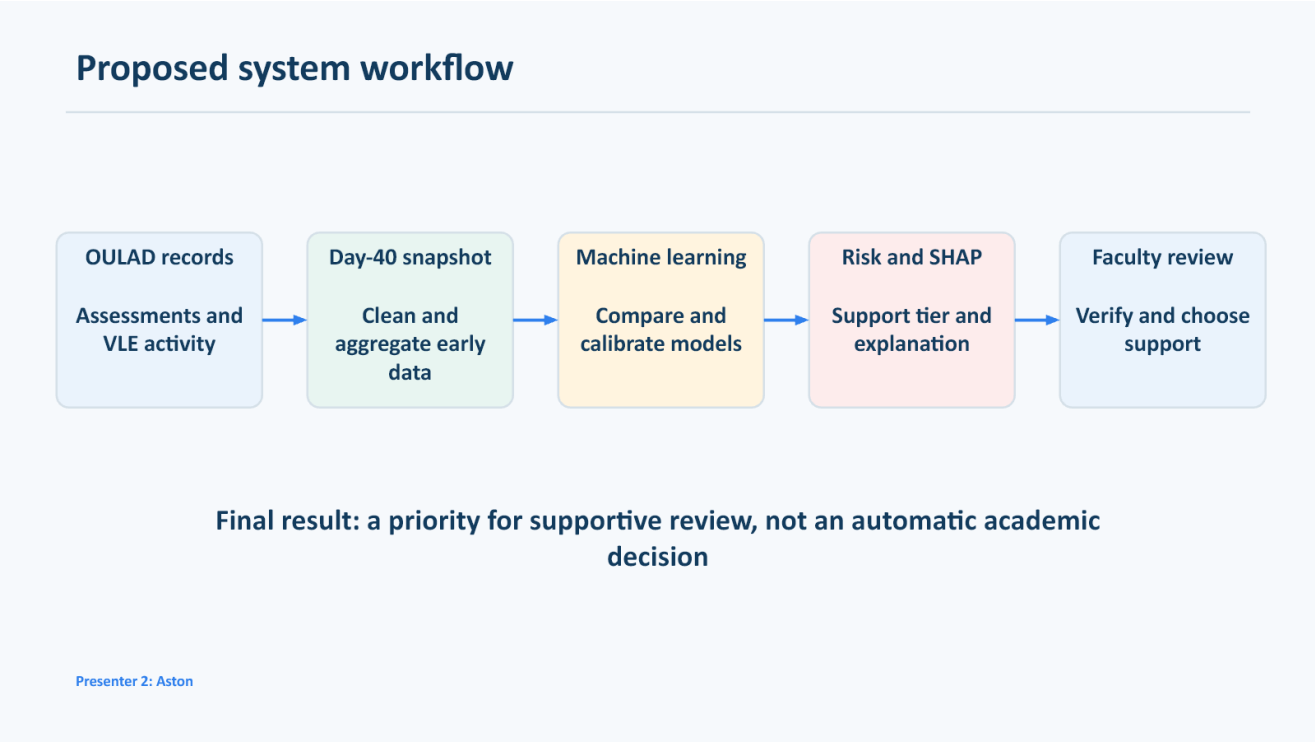


Speaker notes

Our main objective is to identify support needs early and provide understandable evidence for faculty. We use the Open University Learning Analytics Dataset. It contains assessment, registration and Virtual Learning Environment activity. We build a day-40 snapshot, so the model sees only information available at that time. We do not call VLE activity classroom attendance, and the dataset does not contain GPA or backlogs.

Slide 4: Proposed system workflow

Speaker: Aston	Suggested time: 65 seconds



Speaker notes

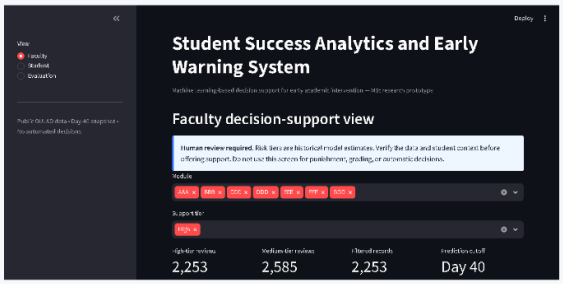
The workflow begins with anonymised OULAD records. We clean the data and build one day-40 record for each student and module. We compare machine-learning models and calibrate the selected model. Its probability becomes a Low, Medium or High support priority. SHAP helps us explain the model output. Faculty then verify the information and decide whether any supportive action is appropriate.

Slide 5: Dashboard views

Speaker: Aston	Suggested time: 60 seconds

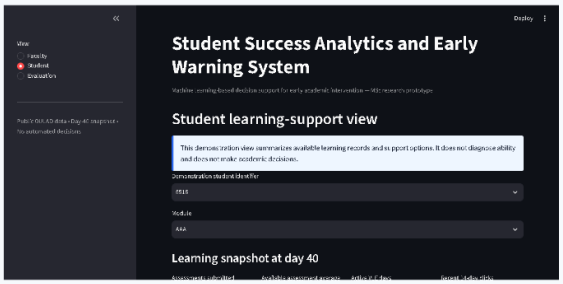
Dashboard views

Faculty view



Shows review priorities, contributing factors and suggested support actions.

Student support view



Shows learning records and support resources without displaying a harmful risk label.

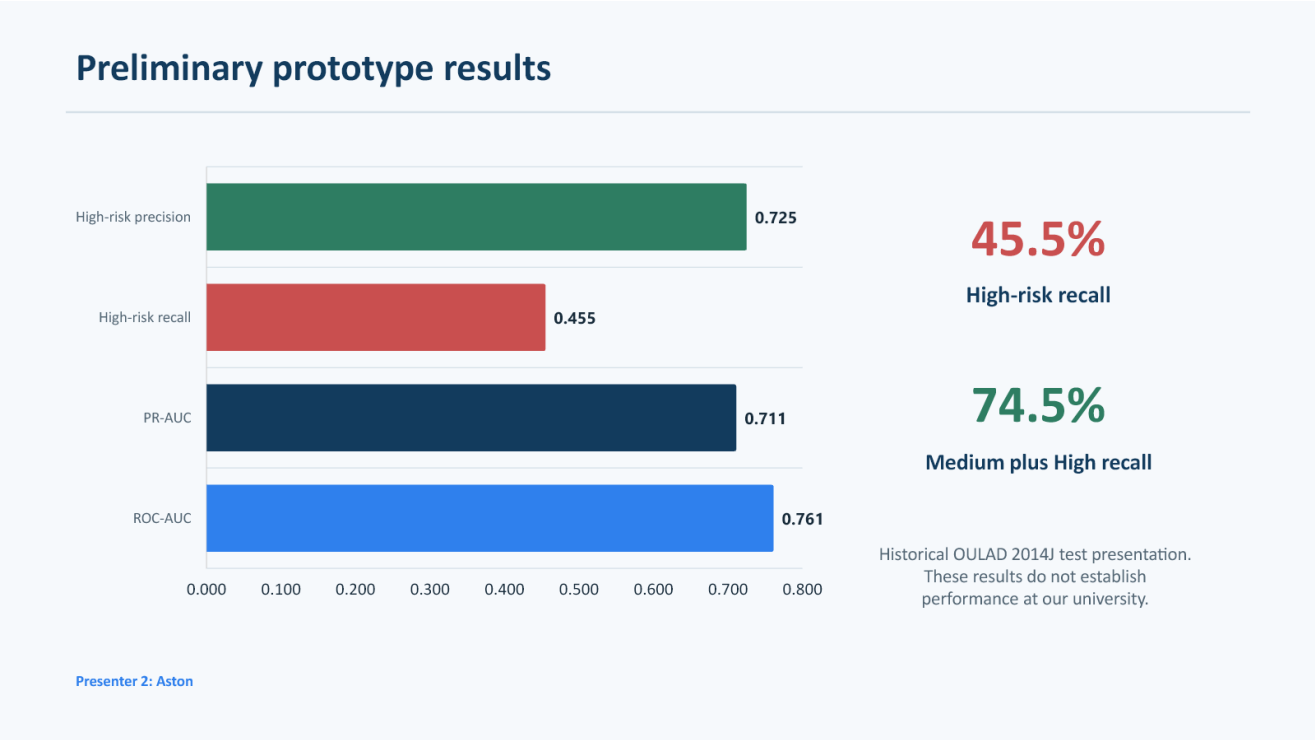
Presenter 2: Aston

Speaker notes

The faculty view organises students by support priority and gives faculty the evidence needed for review. It also displays a clear warning that the system cannot make automatic decisions. The student view focuses on learning records and available support. We intentionally avoid showing a risk label to students because that label could be misunderstood or cause unnecessary anxiety.

Slide 6: Preliminary prototype results

Speaker: Aston	Suggested time: 80 seconds

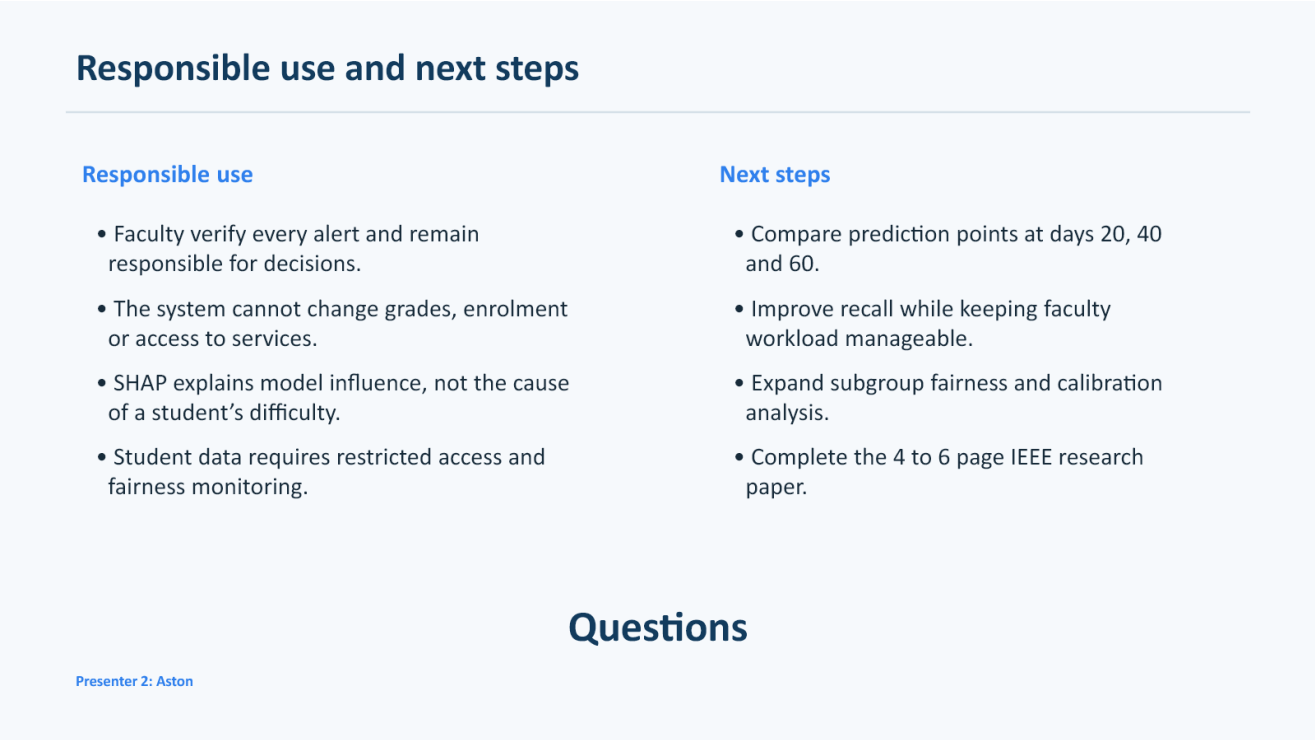


Speaker notes

These are preliminary results from the untouched OULAD 2014J test presentation. The ROC-AUC is 0.761 and the precision-recall AUC is 0.711. High-risk precision is 72.5 percent, but High-risk recall is only 45.5 percent. This means the highest tier still misses many adverse outcomes. When we combine Medium and High, recall increases to 74.5 percent. The findings support further research, but they do not prove that the model will work at our university.

Slide 7: Responsible use and next steps

Speaker: Aston, then both presenters	Suggested time: 65 seconds



Speaker notes

Responsible use forms part of the system design. Faculty must verify every alert, and the system cannot make academic or disciplinary decisions. SHAP explains the model's behaviour but cannot diagnose why a student is struggling. Our next work will compare different prediction points, improve recall, strengthen fairness testing and complete the IEEE paper. In summary, the project aims to help faculty notice possible support needs earlier while keeping people responsible for the final decision. Thank you. We are happy to answer questions.